

CBCS SCHEME

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BCS502

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Computer Networks

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What is data communication? List and explain characteristics and components of communication model.	06	L1	CO1
	b.	Define switching. Explain Circuit Switched Network and Packet Switched Network.	06	L2	CO1
	c.	With neat sketch, explain different layers of TCP/IP protocol suite.	08	L2	CO1
OR					
Q.2	a.	What are guided transmission media? Explain twisted pair cable in detail.	06	L1	CO1
	b.	What is Virtual Circuit Network (VCN)? With neat diagram, explain three phases involved in VCN.	08	L1	CO1
	c.	Write a note on Encapsulation and decapsulation at Source Host for TCP/IP protocol suite.	06	L2	CO1
Module – 2					
Q.3	a.	Define Redundancy. Explain CRC encoder and CRC decoder operation with block diagram.	08	L2	CO2
	b.	Distinguish between Flow Control and Error Control. Explain Stop and Wait Protocol.	08	L2	CO2
	c.	List and explain Control Fields of I-frames, S-frames and U-frames.	04	L2	CO2
OR					
Q.4	a.	What is Hamming distance? With example, explain Parity Check Code.	06	L1	CO2
	b.	Define Framing. Explain character oriented framing and bit-oriented framing.	06	L1	CO2
	c.	With flow diagram, explain CSMA/CA.	08	L2	CO2
Module – 3					
Q.5	a.	Explain virtual-circuit approach to route the packets in packet-switched network.	10	L2	CO3
	b.	Illustrate the working of OSPF and BGP.	10	L3	CO3
OR					
Q.6	a.	Explain IPv6 datagram format.	10	L2	CO3
	b.	Write an Dijkstra's algorithm to compute shortest path through graph.	06	L1	CO3
	c.	Write a note on Routing Information Protocol (RIP) algorithm.	04	L2	CO3
Module – 4					
Q.7	a.	Explain Go-Back-N protocol working.	10	L2	CO4
	b.	With neat sketch, explain three-way handshaking of TCP connection establishment.	10	L2	CO4

OR

Q.8	a.	With an outline, explain selective repeat protocol.	10	L2	CO4
	b.	List and explain various services provided by User Datagram Protocol (UDP).	10	L2	CO4

Module – 5

Q.9	a.	Briefly explain Secure Shell (SSH).	10	L2	CO4
	b.	Write a note on Request message and response message formats of HTTP.	10	L2	CO4

OR

Q.10	a.	With neat diagram, explain the basic model of FTP.	04	L2	CO4
	b.	Describe the architecture of electronic mail (e-mail).	06	L3	CO4
	c.	Briefly explain Recursive Resolution and Iterative Resolution in DNS.	10	L2	CO4

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Module – 1			M	L	C
Q.1	a.	Explain fundamental characteristics and data representation in data communication.	6	L1	CO1
	b.	Discuss types of connection and the basic topologies in networks.	6	L1	CO1
	c.	Explain packet switching and circuit switching with neat diagrams.	8	L1	CO1
OR					
Q.2	a.	Explain the layers of TCP/IP protocol suite.	8	L2	CO2
	b.	Explain types of packet switched networks and evaluate the total delay time of both.	12	L2	CO2
Module – 2					
Q.3	a.	Explain types of errors and Hamming distance. Find the Hamming distance between the following : i) d(000, 011) ii) d(10101, 11110).	8	L3	CO2
	b.	Describe the working of CRC encoder and decoder. Perform division with respect to the following : Data word : 1001 Divisor : 1011.	12	L3	CO2
OR					
Q.4	a.	Explain stop-and-wait protocol with FSM.	6	L2	CO2
	b.	Explain the three types of frames in HDLC.	8	L2	CO2
	c.	Discuss controlled-access protocol using reservation method.	6	L2	CO2
Module – 3					
Q.5	a.	Explain the services offered by network layer.	6	L2	CO2
	b.	Define address space. Differentiate between classful addressing and classless addressing.	8	L2	CO2
	c.	Explain Network Address Resolution (NAT) with a neat diagram.	6	L2	CO3
1 of 2					

OR

Q.6	a.	Explain IPv6 packet format in detail.	6	L2	CO2
	b.	Discuss D-V routing highlighting the importance of distance vector.	7	L2	CO2
	c.	Describe BGP protocol in detail.	7	L2	CO4

Module – 4

Q.7	a.	Explain the concept of port numbers mentioning ICANN ranges.	5	L2	CO3
	b.	Explain Go-Back-N protocol.	9	L2	CO4
	c.	Explain TCP segment format with a neat diagram.	6	L3	CO3

OR

Q.8	a.	Discuss the connection establishment in TCP.	8	L2	CO3
	b.	Explain error control in TCP using acknowledgements.	4	L2	CO3
	c.	Discuss three algorithms for handling congestion in TCP.	8	L2	CO3

Module – 5

Q.9	a.	Discuss application layer paradigms with neat diagram.	5	L2	CO3
	b.	Explain the use of sockets in process-to-process communication.	7	L2	CO3
	c.	Discuss the connection types in HTTP along with formats of messages.	8	L2	CO3

OR

Q.10	a.	Explain POP and IMAP protocols.	8	L2	CO4
	b.	Discuss the applications of SSH protocol.	4	L2	CO4
	c.	Explain resolution in DNS.	8	L2	CO3

Fifth Semester B.E./B.Tech. Degree Examination, June/July 2025

Computer Networks

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Max. Marks: 100

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Module – 1			M	L	C
Q.1	a.	Define Data Communications. Explain the characteristics and components of Data communication with neat diagram.	10	L2	CO1
	b.	With neat diagram explain the Layers in the TCP /IP protocol suite.	10	L2	CO1
OR					
Q.2	a.	Explain in detail the guided and unguided Media transmission with suitable diagram.	12	L2	CO1
	b.	Describe the working of Datagram network with suitable sketches	08	L2	CO1
Module – 2					
Q.3	a.	With a neat sketch describe the working of simple protocol of Data Link Layer. Develop a program to implement a sliding window protocol in the data link layer.	12	L2	CO2
	b.	Illustrate the stop and wait protocol of DLL with an example.	08	L2	CO2
OR					
Q.4	a.	Solve : i) In parity check if the dataword is 1011. What is the code word? What happens at receiver, if the receive word is a) 10011 b) 10110 c)01011 ii) Generate CRC for the dataword $x^3 + 1$ and the generator $x^3 + x + 1$. What happens if the received word is 1000110. iii) Generate checksum of list of five 4-bit number (7,11,12,0,6) and verify the same at receiver.	12	L3	CO2
	b.	Illustrate the working of CSMA/CA with a flow diagram	08	L2	CO2
Module – 3					
Q.5	a.	Summarize the packet format of IPV6 datagram with suitable diagram.	10	L2	CO2
	b.	Develop an algorithm for Distance Vector Routing and explain the same.	10	L2	CO4
OR					
Q.6	a.	Explain MOSPF with an example and suitable diagram.	10	L3	CO4
	b.	Develop algorithm for Link state Routing and explain the same.	10	L2	CO4
Module – 4					
Q.7	a.	Illustrate the working of Go-back-N protocol with an example	12	L2	CO4
	b.	Explain connectionless and connection oriented services in Transport layer.	08	L2	CO2
OR					
Q.8	a.	Illustrate the connection establishment and termination in TCP/IP with suitable sketches.	12	L2	CO3
	b.	With sketch of TCP segment format, describe its field.	08	L2	CO3
Module – 5					
Q.9	a.	Explain FTP and its two connections.	10	L2	CO3
	b.	Explain SMTP with diagram and the mail transfer phases.	10	L2	CO3
OR					
Q.10	a.	Explain MIME and its header.	10	L2	CO3
	b.	Explain SSH and its components with neat diagram.	10	L2	CO3

Module -1		
Q.01	a	Describe the operation of connectionless packet-switched network with the help of neat diagram.
	b	What is meant by logical connection in TCP/IP. Explain TCP/IP reference model with diagram?
	c	Explain the four basic topologies used in networks. List advantages and disadvantages of each of them.
OR		
Q.02	a	What are guided transmission media? Explain twisted pair cable in detail.
	b	Describe TCP/IP reference model with diagram?
	c	Compare OSI and TCP/IP Models. What are the reasons for OSI model to fail?
Module-2		
Q. 03	a	Solve using Cyclic Redundancy Check. Dataword:1001, Divisor-1011.
	b	Write the S Frame format of HDLC and explain the individual field of it.
	c	What is the difference between ALOHA and Slotted ALOHA
OR		
Q.04	a	What is bit oriented framing and its frame pattern. Explain with example byte stuffing and unstuffing in bit-oriented framing.
	b	Explain the flow diagram of CSMA/CD and explain.
	c	What is PPP? What are the services provided by PPP?
Module-3		
Q. 05	a	Explain classful addressing system with a neat diagram.
	b	Show, how to form the least cost tree using Dijkstra Algorithm with an example.
	c	An ISP is granted the block 16.12.64.0/20. The ISP needs to allocate addresses for 8 organizations, each with 256 addresses. a. Find the number and range of addresses in the ISP block. b. Find the range of addresses for each organization and the range of unallocated addresses. c. Show the outline of the address distribution and the forwarding table.
OR		
Q. 06	a	Write a program for Bellman ford algorithm.
	b	Explain Open Shortest Path First Protocol with example.
	c	Explain DHCP and its importance?
Module-4		
Q. 07	a	Explain Transport-Layer Services Process-to-Process Communication in detail

	b	List the services and applications of TCP.
	c	Explain stop and wait and Selective-Repeat protocol working.
OR		
Q. 08	a	Explain connection establishment of TCP using 3-way handshaking.
	b	Explain TCP Congestion control.
	c	Explain FSM for Reno TCP
Module-5		
Q. 09	a	Explain the services provided by the Transport Layer with different protocols
	b	Differentiate client server paradigm and peer-to-peer paradigm.
	c	Differentiate between Persistent and Non-Persistent connection in HTTP.
OR		
Q. 10	a	Explain about Web Documents and HTTP
	b	Explain in detail Iterative Communication Using TCP
	c	Explain briefly Domain Name System (DNS)